BENEATH THE SURFACE

AI-Driven Subsurface Visualization for Public Understanding

Team Members: Rafik, Seid, Vimal, and Sahil
Faculty: Dr. Areti Markopoulou, Dr. Joaquin David Rodríguez Álvarez, Akshay Madapura

THE INVISIBLE CRISIS

40%

Decline in soil organic carbon since 1950

2075

Irreversible compaction threshold

INVISIBLE

SLOW

CUMULATIVE

How can AI-based prediction translate subsurface transformations into experiential, debatable knowledge for All?

What it is?

Beneath the Surface is an interactive installation that uses AI-driven geospatial analysis to visualize historical years of anthropogenic transformation in Barcelona’s soil and water strata.

Why it matters?

Contemporary architecture often ignores the subsurface. By making invisible soil degradation (carbon loss, aquifer compaction, and contamination) experiential, we reframe the ground not as a passive resource, but as a critical infrastructure of collective survival.

AI for All:

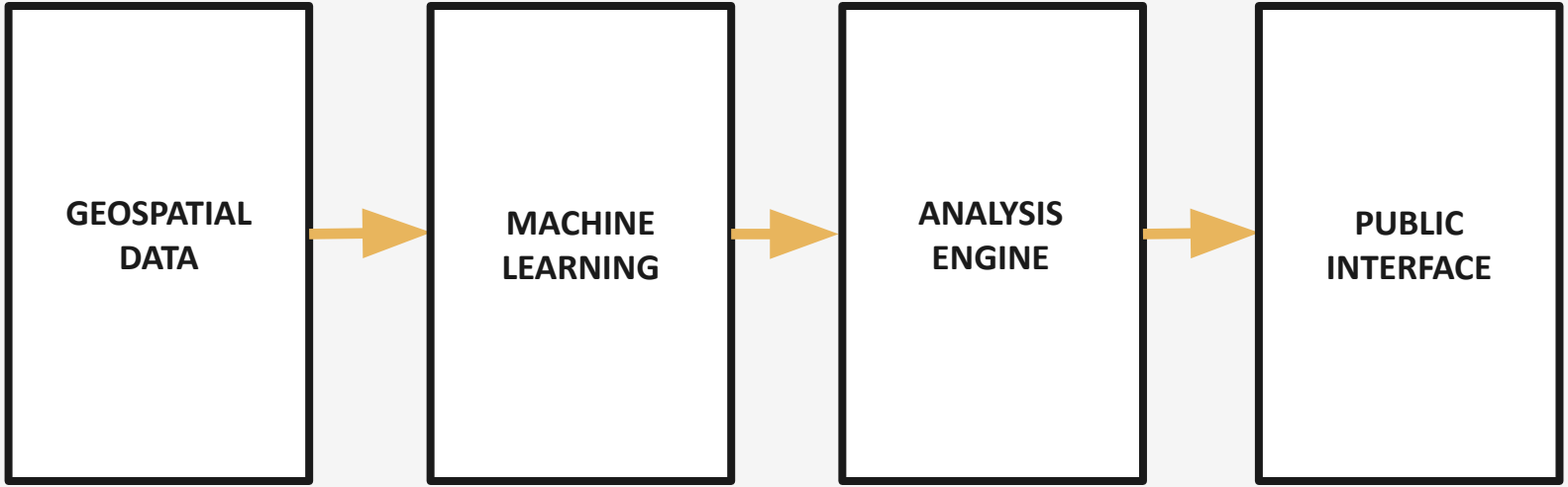
We democratize environmental knowledge by using **AI as a Translator**. It takes cryptic, technical soil data (pH, bulk density, SOC) and converts it into intuitive light-scapes and predictive scenarios, allowing the public to debate and understand long-term ecological evolution.

Typology

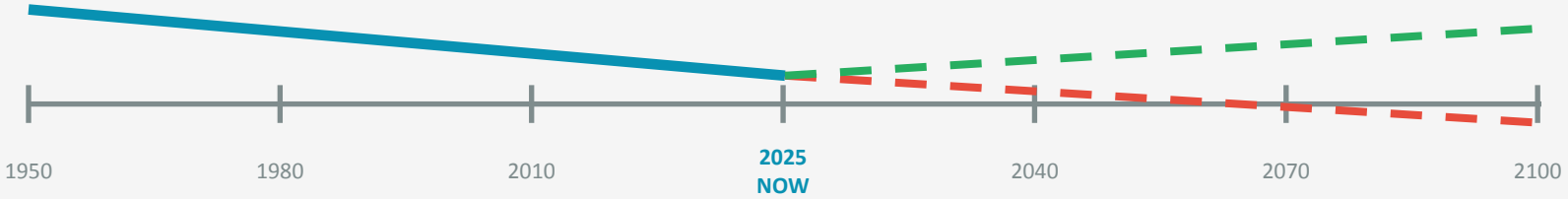
AI-powered Public Awareness Tool & Speculative Storytelling.

The project acts as a diagnostic bridge between historical geological archives and potential urban futures, using AI to narrate the consequences of human intervention.

METHODOLOGY



TEMPORAL INTERPRETATION



Historical Data (1950-2018)

↓ Degradation Scenario

↑ Regenerative Scenario

CURRENT STATE (2025)

Organic carbon: 12 g/kg

pH: 7.2 | Compaction: moderate

INSTALLATION DESIGN

SOIL MONOLITH TABLE

60 samples (10×6 grid)

+ 60" Display Screen

+ Interactive Station

1 **SOIL TABLE**
60 cubes, real samples

2 **SCREEN**
60" LCD, real-time viz

3 **STATION**
Touch interface, scenarios

2.0m × 1.5m × 2.2m

AI AS EPISTEMIC MEDIATOR

Not optimization - translation

TRANSLATES DEEP TIME

REVEALS PATTERNS

PRESERVES UNCERTAINTY

PROJECT TYPOLOGIES

PUBLIC AWARENESS

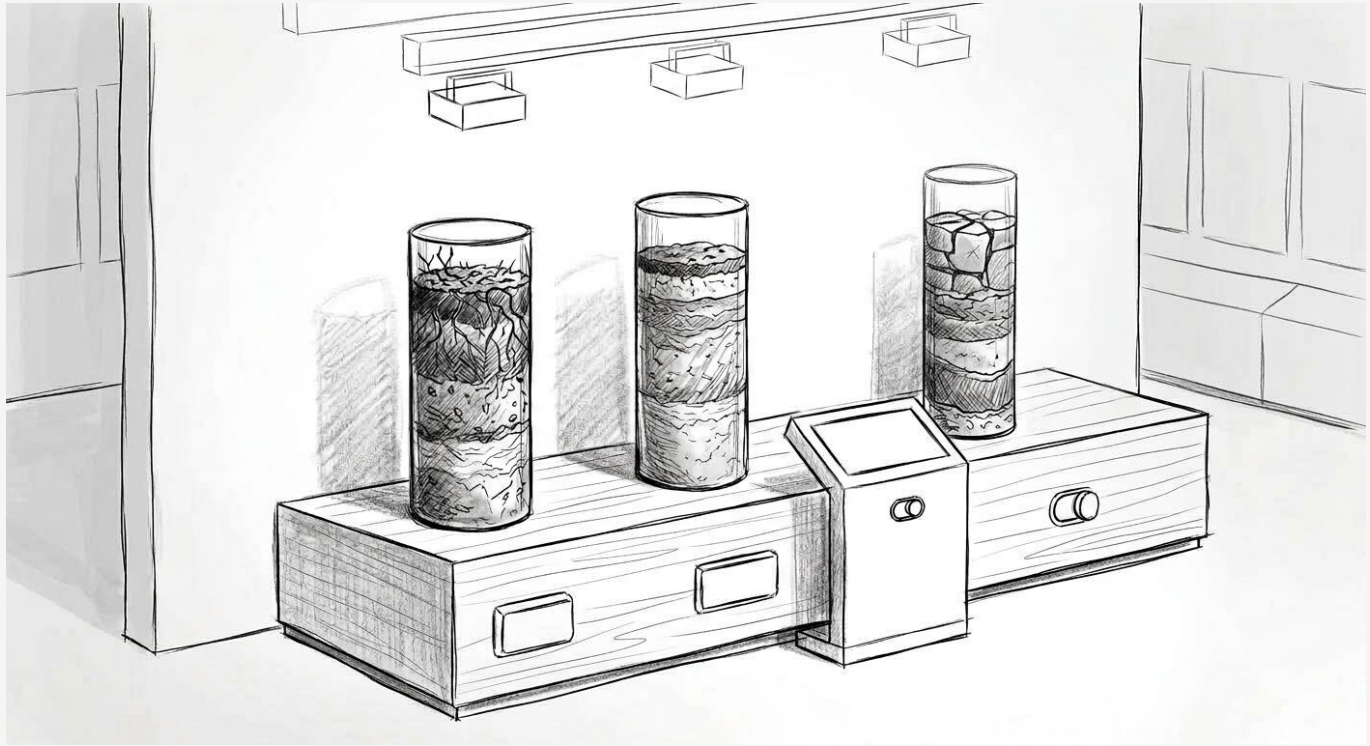
INTERACTIVE EXPERIENCE

CO-DESIGN LOGIC

SPECULATIVE FUTURES

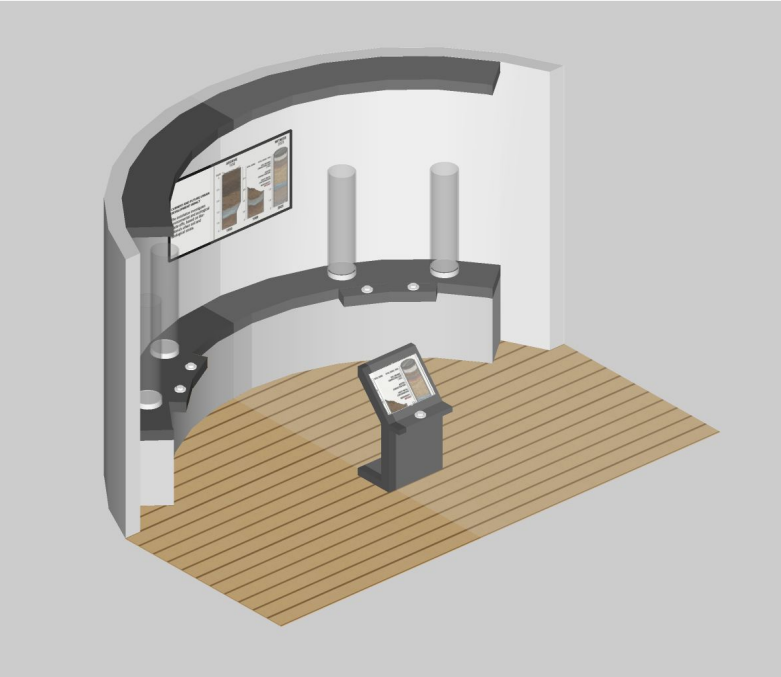
INTERACTION PAVILION

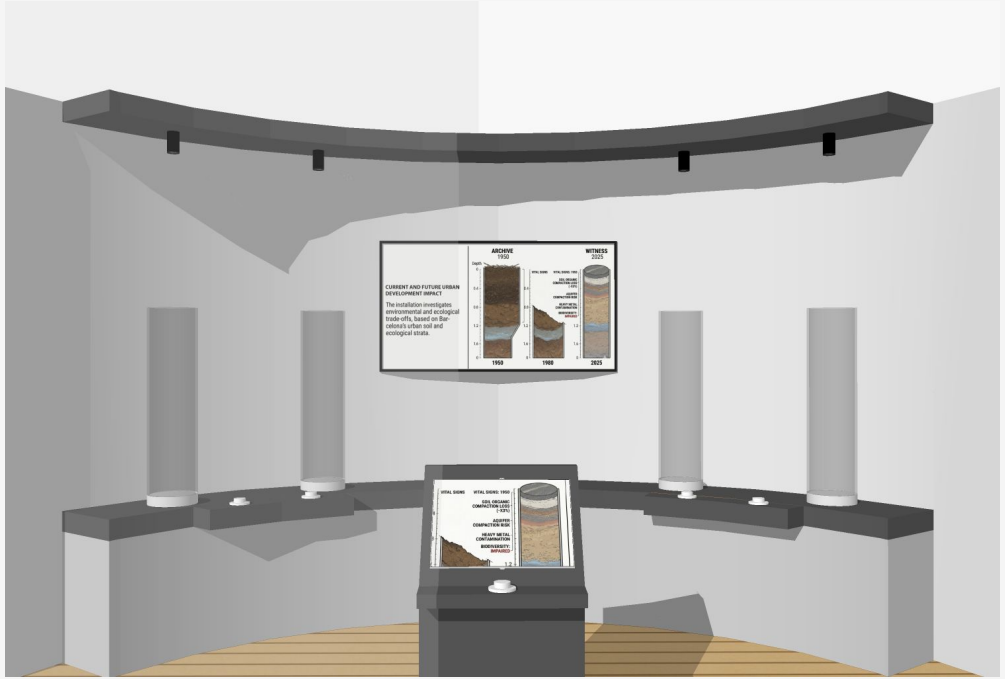
thematic approach



INTERACTION PAVILION

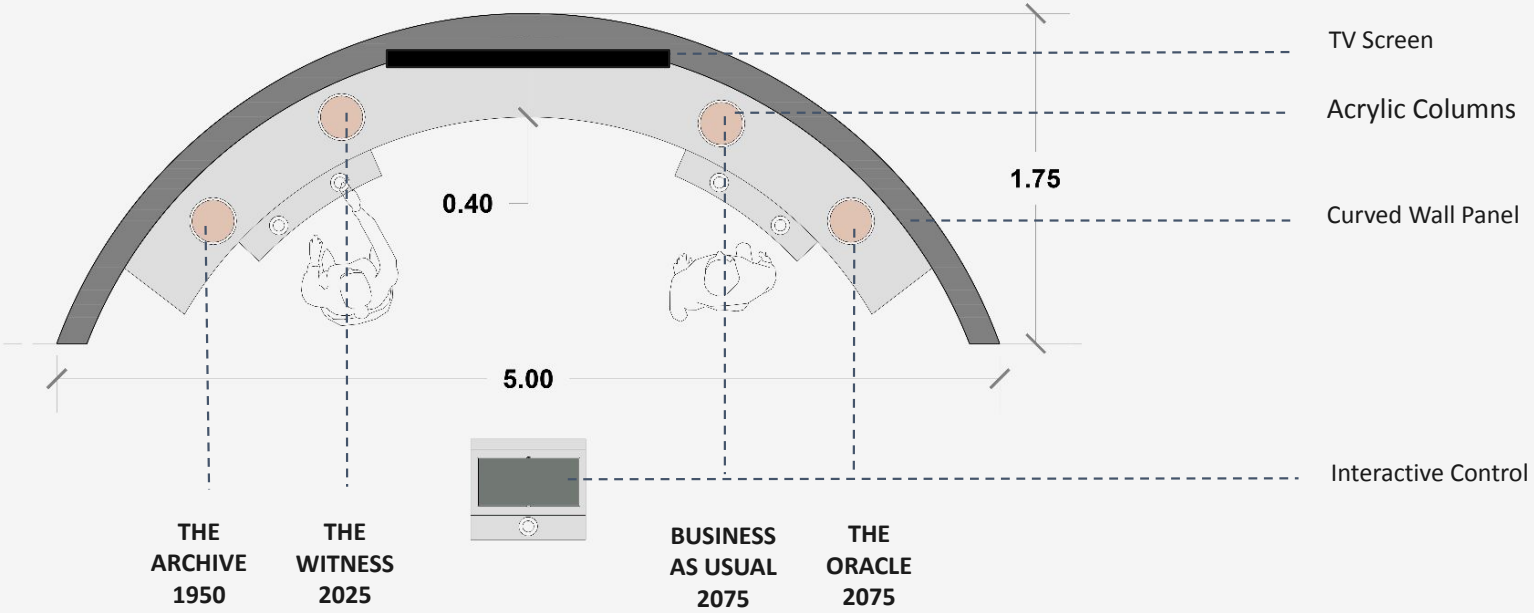
thematic approach





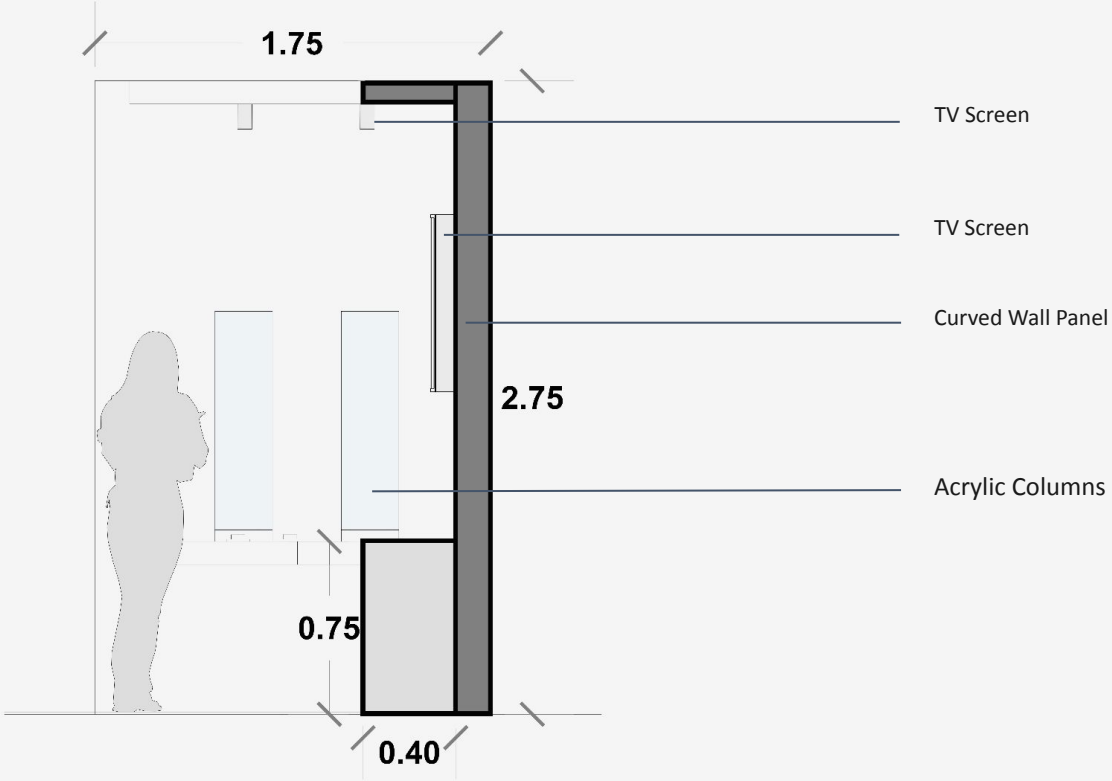
INTERACTION PAVILION

Floor Plan



INTERACTION PAVILION

Section

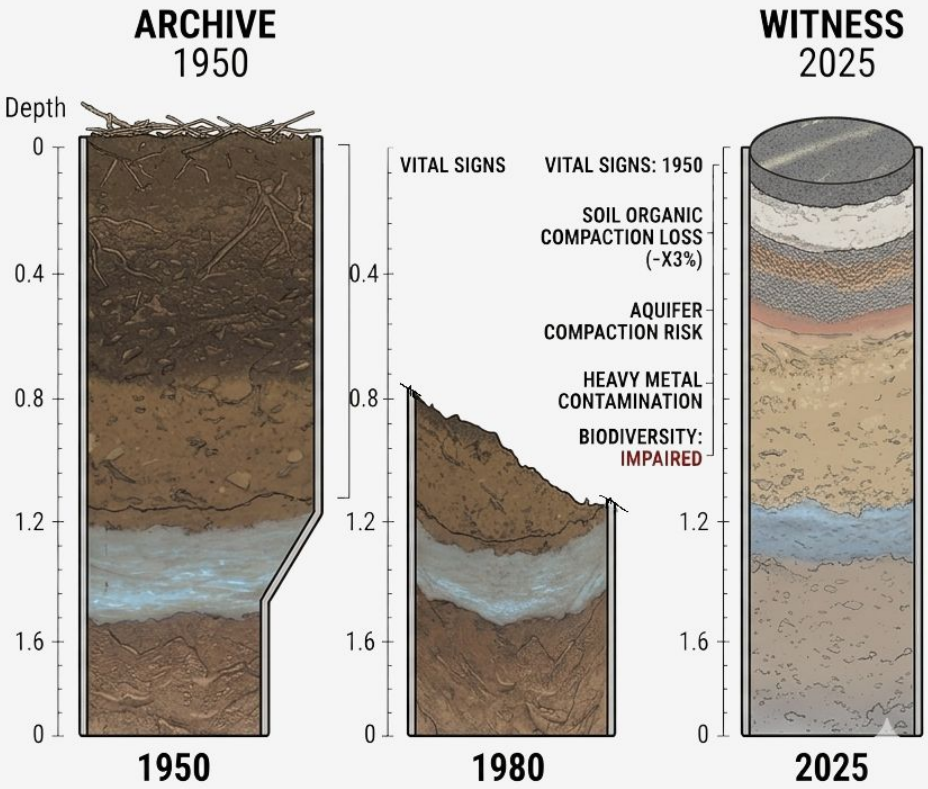


INTERACTION PAVILION

Details

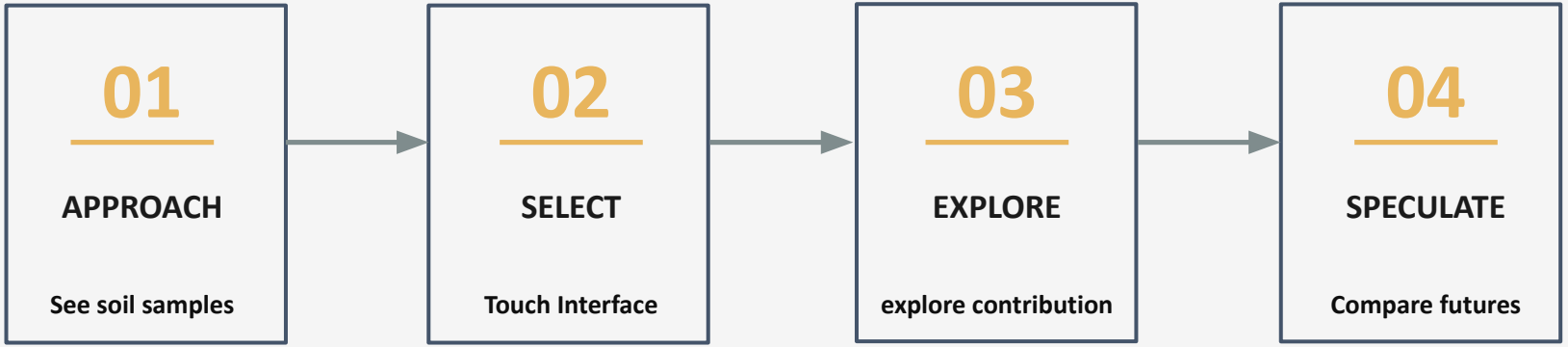
CURRENT AND FUTURE URBAN DEVELOPMENT IMPACT

The installation investigates environmental and ecological trade-offs, based on Barcelona's urban soil and ecological strata.

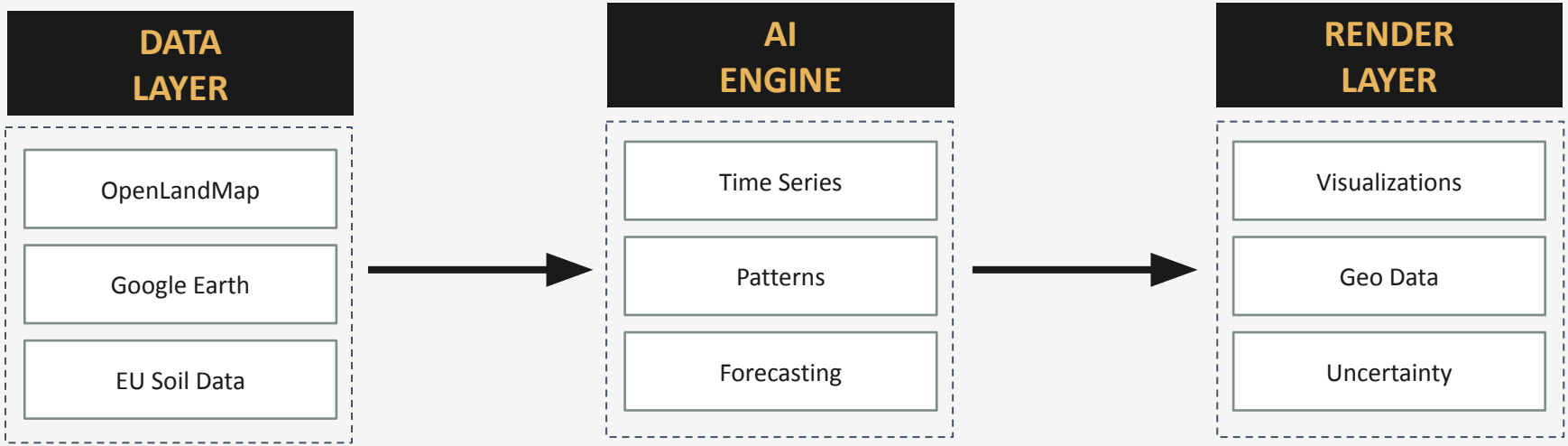


CURRENT AND FUTURE URBAN DEVELOPMENT HAPACT
Tbe installation inveetigats emelronmenial and ocmagical frade-offe, based on Bavcelore's urben coil and ecological etrate.
ARCHIVE
1950
WITNCSS
2025
1950
1960
2025

VISITOR EXPERIENCE



SYSTEM ARCHITECTURE



PREDICTING THE FUTURE

From real measurements to 50-year forecasts

INPUTS

Real Satellite Data (2020)
- Organic Carbon 12.4 g/kg
- Soil pH 6.8
- Clay Content 28%
- Sand Content 41%
- Bulk Density 1.3 t/m³

Climate Trajectory (IPCC AR6)
- Temperature +2.7°C by 2075
- Rainfall −20% by 2075
- CO_2 600 ppm by 2100

Land Management Choice
- Rewild / Farm / Build

THE AI ENGINE

RothC Carbon Model

Physics-based carbon decomposition model (IPCC-endorsed, 40yr)

+

Random Forest Classifier

Spatial pattern recognition across soil properties

93%

test accuracy

215,660 pixels

trained on real Spain GeoTIFF data

OUTPUTS

Per depth layer, per year
- Predicted carbon value
- Water Retention (%)
- Bulk Density (t/m³)
- Confidence range ±
- % change from today

For 50-year horizon
- Annual values 2025–2075
- 10 ensemble runs
- P10/P90 bounds

For the exhibition
- Column fill height
- Colour intensity
- Warning thresholds

93% accuracy on held-out test data · <3 seconds per simulation · 400 grid cells × 50 years · 10 ensemble members

THE DATA: WHERE IT COMES FROM

The Data Behind Every Prediction

Soil Data

- OpenLandMap / SoilGrids — ISRIC, Netherlands
- Fused satellite imagery + 150,000 field samples globally
- 6 properties measured at 6 depth bands (0–200cm)
- Spain coverage: 250m resolution GeoTIFF rasters

Properties:

Organic Carbon · Clay · Sand · Bulk Density · pH · CEC

Climate Data

- CHIRPS precipitation (2000–2024) — NASA/UCSB
- MODIS land cover (2001–2023) — NASA Terra satellite
- IPCC AR6 temperature projections — calibrated to Barcelona baseline (16.2°C, 580mm/yr)

Why these datasets?

- Publicly available, globally validated, reproducible
- The 6 soil properties are the exact inputs required by the RothC physics model — not a choice, a necessity
- CHIRPS and MODIS cover the full 23-year historical training window

THE MODEL: WHAT IS IPCC AND WHAT ARE THE KEY ASSUMPTIONS MADE

What the Model Assumes — and Why

What is IPCC?

- UN body synthesising climate science from 195 countries, 10,000+ researchers
- AR6 (2021): most comprehensive climate assessment ever published
- SSP scenarios = Shared Socioeconomic Pathways — 4 futures based on emissions choices
  - SSP1: +1.3°C (fast action) → SSP5: +5.0°C (worst case)
- Used by every government, every climate model, every adaptation plan globally

The 5 Land Management Assumptions

STRATEGY	KEY ASSUMPTION
Let Nature Recover	Natural succession, no human input, carbon rises slowly over decades
Traditional Farming	Current Spanish avg: 20cm tillage depth, 80kg/ha synthetic nitrogen
Agroforestry	30% tree canopy within 10 years, erosion −40%, litter increases humus
Intensive Regenerative	2.5 t/ha/yr organic input, no-till, cover crops — optimistic ceiling
Precision Sustainable	Sensor-matched inputs, deficit irrigation — realistic technology adoption

The model predicts what soil would do if these practices applied uniformly.
It cannot model farmer behaviour, land ownership, subsidies, or political will.

THE ALGORITHM: ACADEMIC VALIDATION

The Models Are Not New — They Are Proven

Used by the IPCC, the FAO, and soil scientists on six continents

RothC Carbon Model

Developed: Rothamsted Research, UK — Coleman & Jenkinson, 1996

Type: Mechanistic soil carbon turnover model

Core logic: Models 5 carbon pools (DPM, RPM, BIO, HUM, IOM) and their decomposition rates driven by temperature, moisture, and clay content

Used by:

- IPCC Working Group I — AR5 and AR6 land carbon assessments
- FAO Global Soil Carbon Assessment (2022)
- European Environment Agency — soil carbon monitoring
- UK Centre for Ecology & Hydrology — national carbon stock modelling

Tested in: UK, Australia, Brazil, Kenya, China, India

Peer-reviewed validation:

- *Coleman, K. & Jenkinson, D.S. (1996). RothC-26.3 — A model for the turnover of carbon in soil. Evaluation of Soil Organic Matter Models, Springer, pp. 237–246.*
- *Falloon, P. & Smith, P. (2002). Simulating SOC changes in long-term experiments. Soil Use and Management, 18(2), 101–111.*
- *Gottschalk, P. et al. (2012). How will organic carbon stocks in mineral soils evolve under future climate? Global Change Biology, 18(5), 1765–1784.*

Pedotransfer function (Water Retention):

Hudson, B.D. (1994). Soil organic matter and available water capacity. Journal of Soil and Water Conservation, 49(2), 189–194.

Random Forest Classifier

Developed: Breiman, 2001 — UC Berkeley

Type: Ensemble of decision trees, majority-vote classification

Our deployment: 215,660 pixels, 5 features, 93.03% accuracy

Used in soil science for:

- SoilGrids global soil mapping (ISRIC) — the exact dataset we trained on
- ESDAC European soil degradation mapping
- Predicting soil organic carbon across national scales (FAO, 2020)
- Land degradation classification — UN Convention to Combat Desertification

Peer-reviewed validation:

- *Breiman, L. (2001). Random Forests. Machine Learning, 45(1), 5–32. [25,000+ citations]*
- *Hengl, T. et al. (2017). SoilGrids250m: Global gridded soil information based on machine learning. PLOS ONE, 12(2), e0169748.*
- *Lamichhane, S. et al. (2019). Digital soil mapping algorithms and covariates for soil organic carbon mapping. Geoderma, 352, 323–338.*

Our Model's Accuracy

Training samples
215,660 spatial pixels

Cross-validation accuracy
93.03%

Features used
OC, clay, bulk density, precip, land cover

Validation method
Stratified k-fold on held-out Spain tiles

Temporal coverage
2001–2023 (23 years)

"RothC has been tested against measured data from long-term experiments on five continents. It is one of the most validated soil carbon models in existence."

— Smith et al., 1997, Global Change Biology

WHY YOU CAN TRUST THIS PREDICTION

Three reasons — not one

REAL DATA

NOT ASSUMPTIONS

215,660

real pixel measurements

Trained on real soil measurements across Spain. 93% accuracy on soil texture classification. Source: OpenLandMap / SoilGrids satellite data. Not simulated. Not estimated. Measured from the ground.

REAL PHYSICS

NOT JUST STATISTICS

40

years of validated science

RothC is used by the IPCC itself to model carbon in soil. It knows how carbon decomposes. How clay protects it. How heat accelerates its loss. the model runs the physics, not just the patterns.

REAL UNCERTAINTY

NOT FALSE CERTAINTY

74%

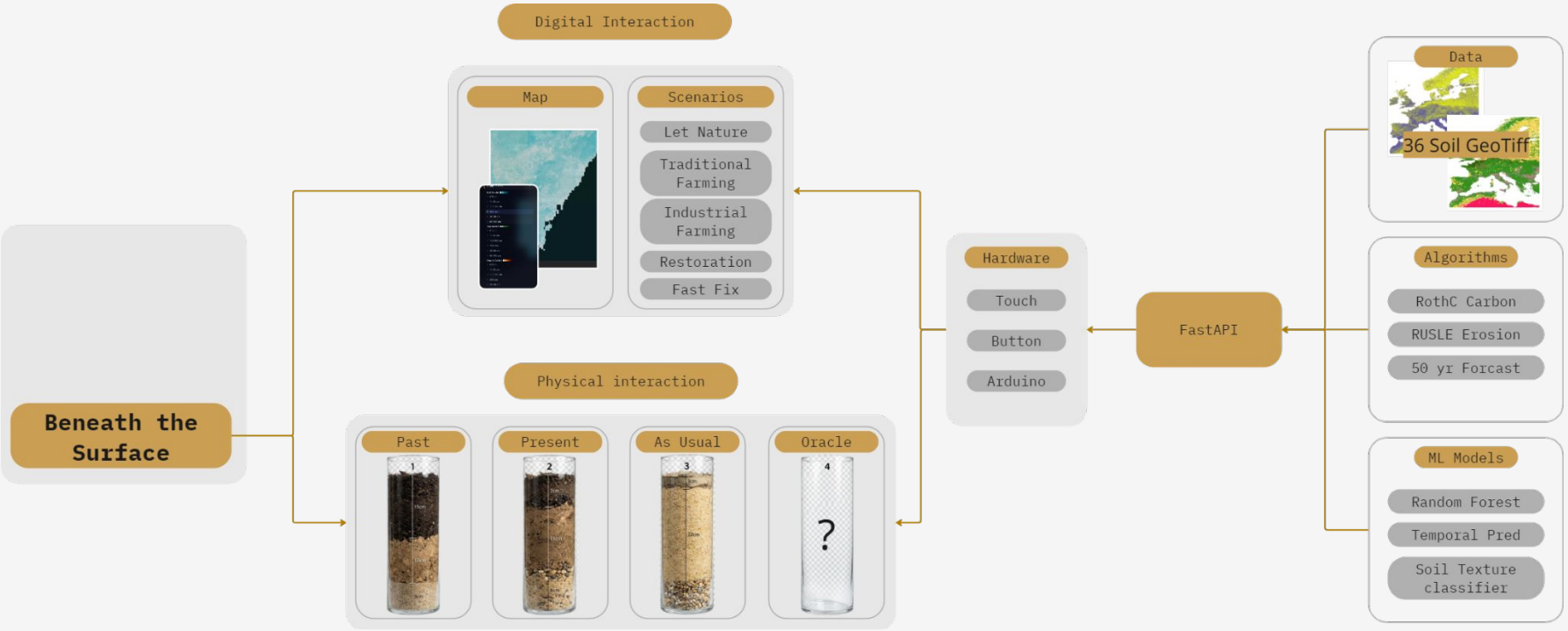
confidence shown, not hidden

Every prediction shows a confidence range: '4.1 g/kg likely between 2.8–6.1'. The model tells you what it doesn't know. That is the honest part.

A scientist would need 50 years of fieldwork to see what this model shows in 8 seconds.
The difference is not certainty — it is speed. The uncertainty is the same. We just make it visible.

TRANSLATING THE SYSTEM INTO THE PAVILION

System Architecture diagram



PROJECT BUDGET

4 Interactive Columns with AI-Powered Fill System

COST BREAKDOWN

Physical Installation	**€490**
• 4 Acrylic Columns (30cm ø × 1m H)	€240
• Colored Soil (60kg)	€120
• Curved Wall Structure	€80
• Mounting Hardware	€50
Electronics & Control	**€268**
• Arduino + 8 Servos	€104
• Touch Screen Kiosk	€120

TOTAL COST **€638**

FUNDING STRATEGY

Base Budget Available	**€500**
Cost Optimizations:	
• IAAC Fab Lab Support	**-€100**
• Compact Design (30cm × 1m)	
• Team Laptop (Provided)	**-€0**
• Venue Provides Flooring	**-€0**

ADDITIONAL NEEDED **+€38**

(Minimal Additional Support)

DETAILED BUDGET

Component-by-Component Analysis

Component	Specification	Qty	Unit	Total
Acrylic Cylinders	30cm ø × 100cm H, 5mm	4	€60	**€240**
Acrylic Bases/Lids	30cm ø discs, 8mm	8	€8	**€64**
Colored Sand/Soil	6 colors, 15kg each	60kg	€2/kg	**€120**
Curved Wall Panel	Plywood, 3m span	1	€80	**€80**
Platform/Mounting	Brackets, pedestals	1 set	€50	**€50**
Arduino Mega	2560 R3, 16 PWM pins	1	€40	**€40**
Servo Motors	MG996R, 20kg torque	8	€8	**€64**
TV Screen	From IAAC	1	€0	**€0**
LED Strips	RGB 5050, 5m	2	€15	**€30**
Power Supply	12V 5A + 5V 3A	1	€14	**€14**

**** Team laptop provided | Venue flooring | IAAC Fab Lab discount: - €200***

Physical Installation: **€554**

Electronics & Control: **€148**

TOTAL PROJECT COST **€638**

BUDGET FEASIBILITY

Implementation Options Comparison

PROPOSED (30cm × 1m)

- 4 Columns
- Compact size
- Interactive Touch Screen
- TV Screen
- Full LED system

€638

+€38
Nearly Within Budget

SCALED (3 Columns)

- 3 Columns
- 30cm × 1m
- Interactive Touch Screen
- TV Screen
- Reduced LEDs

€468

+€0 (IAAC)
Fully Within Budget

MINIMAL (Manual)

- 2 Columns
- 25cm × 80cm
- Interactive Touch Screen
- TV Screen
- No automation

€420

Under Budget
Basic Implementation

Proposed Design: Only €38 additional needed with IAAC support (€638 total with support)

THANK YOU

BENEATH THE SURFACE

Making the invisible ground

legible, contestable, debatable

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